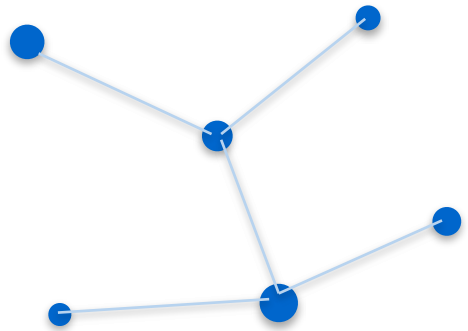
PS-003

Federated Traffic-Flow Prediction with Communication-Cost Evaluation



Privacy-aware collaborative learning for traffic forecasting

TEAM RuminateX

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AIORI Hackathon 3

Abstract

Project summary and intended contribution

Overview

Traffic-flow forecasting can help anticipate congestion, but centralized data collection may require moving sensitive traffic records across locations. This project proposes a federated learning workflow in which multiple traffic nodes train locally and share model updates with a coordinating server.

The prototype will evaluate prediction quality alongside communication overhead, rather than treating accuracy as the only success measure.

Core idea

Keep raw traffic data at each node; exchange model updates for collaborative training.

What we will measure

- Prediction error: MAE and RMSE
- Communication: bytes sent and rounds
- Runtime: training time and convergence

Background

Why federated learning and communication cost matter

Traffic forecasting

Historical traffic speed, volume, and density can be used to estimate future traffic conditions.

Federated learning

Local participants train on their own data and send model updates for aggregation; raw records need not be centrally collected.

Communication overhead

Repeated model exchanges consume bandwidth and time. A useful system measures this cost alongside predictive performance.

Literature & RFC Review

Relevant research and implementation resources

[1] McMahan et al. (2017)

Communication-Efficient Learning of Deep Networks from Decentralized Data. Introduces Federated Averaging (FedAvg). arxiv.org/abs/1602.05629

[2] Li et al. (2020)

Federated Optimization in Heterogeneous Networks (FedProx). Discusses training across clients with differing data and resources. arxiv.org/abs/1812.06127

[3] DCRNN / METR-LA

A widely used traffic forecasting benchmark and model reference for spatiotemporal traffic data. github.com/liyaguang/DCRNN

[4] Flower Framework

Open-source framework for implementing federated learning simulations and deployments. flower.ai/docs

Proposed Solution

End-to-end federated traffic-flow prediction workflow

1. Prepare data

Clean traffic features; split data across simulated locations.

2. Train locally

Each client trains a model on its own local traffic records.

3. Send updates

Clients transmit model parameters/updates—not raw traffic records.

4. Aggregate

Server combines updates (e.g., FedAvg) into a global model.

5. Predict & evaluate

Clients use the global model; report forecast error and communication cost.

Inputs

Traffic speed / volume / occupancy, timestamps, and train-validation-test splits.

Outputs

Traffic-flow forecast, MAE/RMSE, bytes transmitted, communication rounds, and runtime.

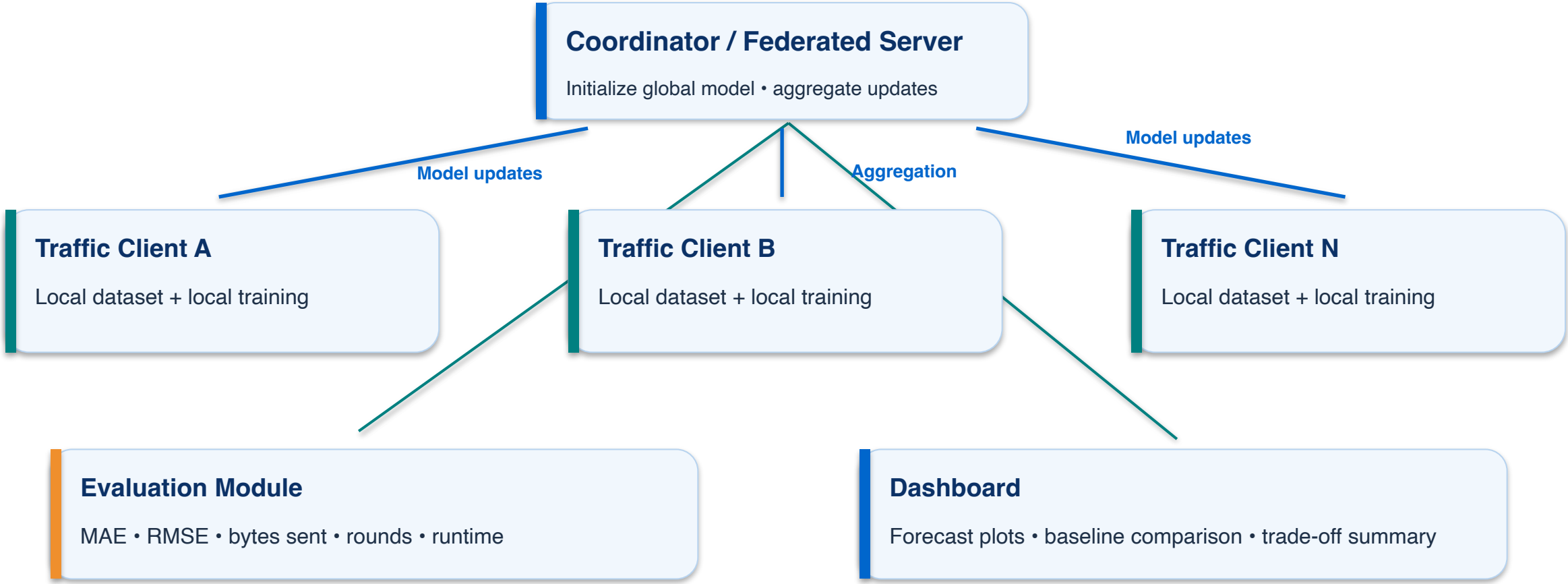
Optimization Proposed

Compare the baseline with candidate communication optimizations

Area	Before / baseline	Proposed test	Evaluation focus
Frequent updates	Baseline: send updates every round	Test fewer communication rounds	Bytes and rounds may decrease; accuracy must be checked
Full-size updates	Transmit full model updates	Test supported compression	Lower payload size; possible accuracy/compute trade-off
Fixed participation	Use same client schedule	Test partial client participation	Potentially lower network load; may affect convergence
Accuracy-only view	Report MAE/RMSE only	Report accuracy + bytes + runtime	Shows practical trade-offs rather than one metric

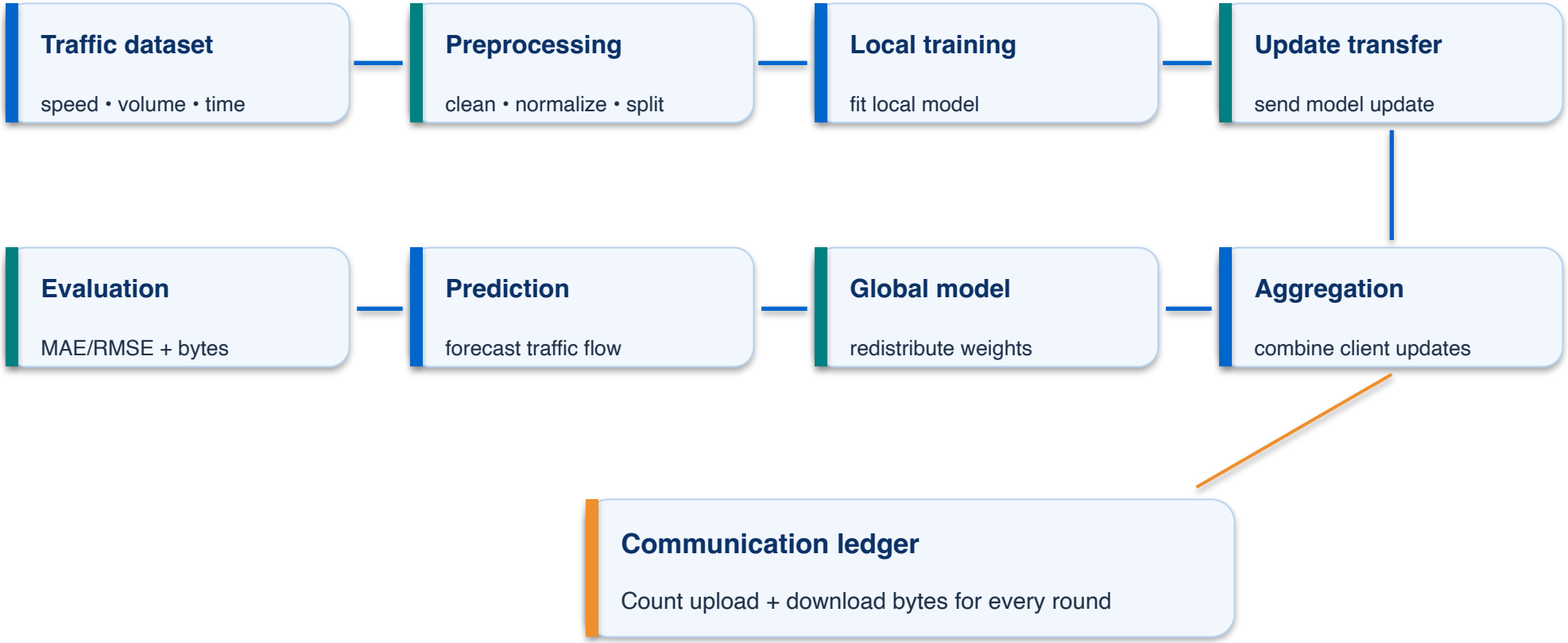
Solution Architecture & Design

Component view of the proposed prototype



Data Flow Diagram

Data and model-update movement through the system



Deployment Timeline

Milestones from research to final October submission

1	Research & scope	Confirm dataset, task, metrics, and baseline.
2	Data preparation	Clean data; split into client datasets; document setup.
3	Baseline model	Train and evaluate a simple traffic prediction model.
4	Federated training	Implement local clients and server aggregation.
5	Cost optimization	Compare update frequency / compression settings.
6	Dashboard & demo	Visualize forecasts, error, bytes, and final findings.

Bibliography

Selected research papers, datasets, and technical documentation

[1] McMahan, B. et al. (2017).
Communication-Efficient Learning of Deep Networks from Decentralized Data.

[2] Li, T. et al. (2020).
Federated Optimization in Heterogeneous Networks (FedProx).

[3] Li, Y. et al. (2018).
Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting.

[4] Flower Documentation.
Federated learning framework documentation.

[5] PyTorch Documentation.
Deep learning framework documentation.

[6] Scikit-learn Documentation.
Preprocessing and evaluation utilities.

Thank You!

RuminateX

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